

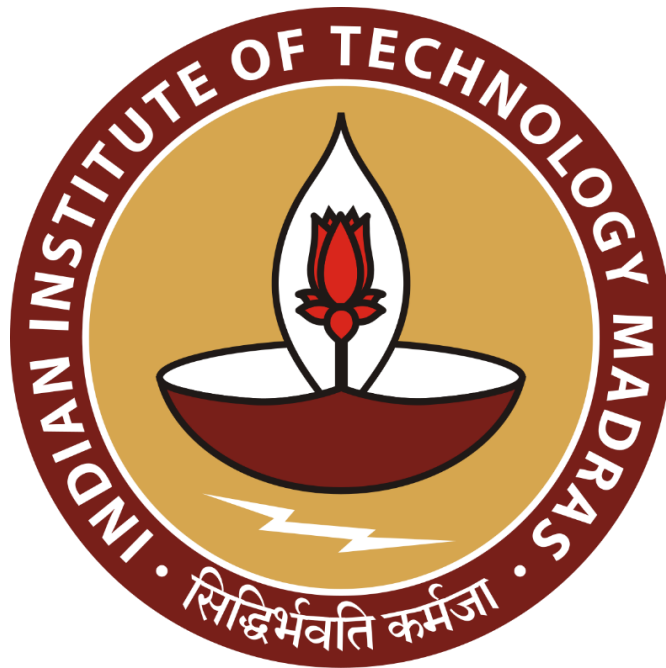
Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title **Sales Trend and Inventory Optimization Analysis of Anand Pharma, Darbhanga**. I extend my appreciation to **Anand Pharma, Donar, Darbhanga, Bihar**, for providing the necessary resources that enabled me to conduct my project.

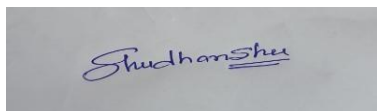
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Shudhanshu' with a stylized underline.

Signature of Candidate

Name: Shudhanshu Kumar Yadav

Date: 12/06/2026

1 Executive Summary

Anand Pharma is a retail B2C pharmacy store operated by Mr. Krishan Mohan Yadav and located in Donar, Darbhanga, Bihar. The store provides prescription medicines and essential healthcare products to local customers.

Anand pharmacy encounters inventory-related challenges, including seasonal demand fluctuations and unstructured restocking decisions. As a result, it experiences overstocking, stock shortages, and financial inefficiencies due to expired or slow-moving medicines.

This project aims to enhance Anand Pharma's inventory efficiency through structured primary data collection and systematic sales pattern analysis. The study will apply descriptive statistics, trend analysis, and product classification techniques to identify fast-moving and slow-moving medicines. The objective is to reduce expiry-related losses, improve stock planning, and support informed, data-driven purchasing decisions for better operational performance.

2 Organization Background

Anand Pharma is a local retail pharmacy located in Donar, Darbhanga, Bihar and operated by Mr. Krishan Mohan Yadav. The pharmacy aims to provide affordable and timely access to essential needed medicines and healthcare products to local residents. Its primary focus is to ensure availability of prescription medicines, general medicines, and basic healthcare items to the local community.

The pharmacy keeps different types of medicines such as tablets, syrups, injections, and other basic health products required in daily life. Because it is located in a residential area, many nearby families depend on it for quick access to medicines without travelling far to bigger hospitals or medical stores. In this way, Anand Pharma plays an important role in meeting the regular health needs of the local community.

With regular customers and steady demand, the pharmacy has been running smoothly for years. However, problems like manual stock management, lack of digital systems, and increasing competition from larger medical stores are slowly affecting its efficiency. Managing expiry dates and maintaining proper stock levels has become more challenging. Therefore, improving inventory management is important for the pharmacy to continue growing and serving customers effectively.

3 Problem Statement

3.1 Inventory Management Issues

Anand Pharma does not follow a structured inventory system. Stock decisions are mostly based on past experience. Because of this, some medicines remain unsold for a long time, while important medicines sometimes go out of stock. This affects both sales and customer trust.

3.2 Seasonal Demand Changes

The demand for certain medicines increases during specific seasons, such as fever and cold medicines during monsoon or winter. However, due to the absence of proper sales analysis, the pharmacy sometimes keeps extra stock or fails to stock enough medicines. This leads to expiry losses or missed sales.

3.3 Demand Forecasting Challenges

Anand Pharma currently does not use any structured demand forecasting method. Purchase decisions are largely based on experience and intuition. As a result, sudden increases or decreases in medicine demand are difficult to predict, leading to stockouts, overstocking, and reduced operational inefficiencies.

4 Background of the Problem

Anand Pharma is a retail B2C pharmacy store operated by Mr. Krishan Mohan Yadav and located in Donar, Darbhanga, Bihar. It deals with many types of medicines, each having different demand levels and expiry periods. The pharmacy serves local residents whose medicine needs change depending on weather conditions and common seasonal illnesses.

Since there is no structured inventory management system, stock decisions are mostly based on experience and manual records. This sometimes leads to overstocking of slow-moving medicines, which increases the risk of expiry and financial loss. At other times, important medicines go out of stock, which affects customer satisfaction and sales.

The problem becomes more serious during seasonal disease outbreaks such as cold, flu, or infections, when demand suddenly increases. Without proper analysis of past sales data, the pharmacy may not be fully prepared for these changes. In addition, limited use of digital tools makes it difficult to track sales patterns and manage inventory efficiently. Because of these issues, there is a need for a more systematic and data-based approach to improve stock management and overall performance.

5 Problem Solving Approach

To address the inventory and demand-related challenges at Anand Pharma, a structured and practical approach will be followed. The main objective is to improve stock management, reduce expiry losses, and ensure that essential medicines are always available for customers.

5.1 Data Collection and Understanding Current System

The first step will be collecting primary sales data from Anand Pharma for a period of 2–6 months. The data will include medicine name, quantity sold, date of sale, purchase price, selling price, and expiry date. Information about current stock levels and supplier lead time will also be noted. Along with numerical data, observations about seasonal illness patterns and customer demand changes will be recorded. All this information will be organized in a simple Excel sheet for further analysis.

5.2 Data Cleaning and Preparation

Before analysis, the collected data will be checked for errors such as missing values, duplicate entries, or incorrect quantities. Medicine names will be standardized to avoid confusion. Proper data cleaning will ensure that the analysis gives accurate and reliable results.

5.3 Sales Trend and Inventory Analysis

Descriptive statistics will be used to calculate total sales, average monthly sales, and revenue contribution of each medicine. Based on sales frequency, medicines will be classified as fast-moving, moderate-moving, or slow-moving. This will help identify which products require regular restocking and which products should be ordered in smaller quantities. Monthly sales trends will also be analyzed to identify seasonal demand patterns. For example, higher sales of fever or cold medicines during certain months will be observed. Simple charts and graphs in Excel will be used to visualize these trends.

5.3.1 ABC Analysis

ABC Analysis will be used to classify medicines based on their annual sales value contribution. Category A medicines will represent high-value medicines contributing the majority of revenue, Category B medicines will represent moderate-value items, and Category C medicines will represent low-value medicines. This classification will help prioritize inventory control and purchasing decisions.

5.3.2 FSN Analysis

FSN (Fast, Slow, Non-Moving) Analysis will be used to classify medicines based on sales frequency and inventory movement. Fast-moving medicines will require frequent replenishment, while slow-moving and non-moving medicines will be monitored to reduce inventory holding costs and expiry-related losses.

5.3.3 Pareto Analysis

Pareto Analysis (80/20 Rule) will be applied to identify the medicines contributing to the majority of sales revenue. This analysis will help focus inventory planning efforts on the most important products and improve overall stock management efficiency.

5.4 Expiry Risk and Reorder Planning

Expiry dates will be compared with sales movement to identify medicines at risk of expiry. Recommendations will be provided to reduce losses, such as adjusting order quantity or promoting slow-moving items. Minimum stock levels will be defined for essential medicines to avoid stockouts.

5.4.1 EOQ Analysis

Economic Order Quantity (EOQ) analysis will be used to estimate the optimal order quantity for selected medicines. The objective is to minimize total inventory costs by balancing ordering costs and inventory holding costs.

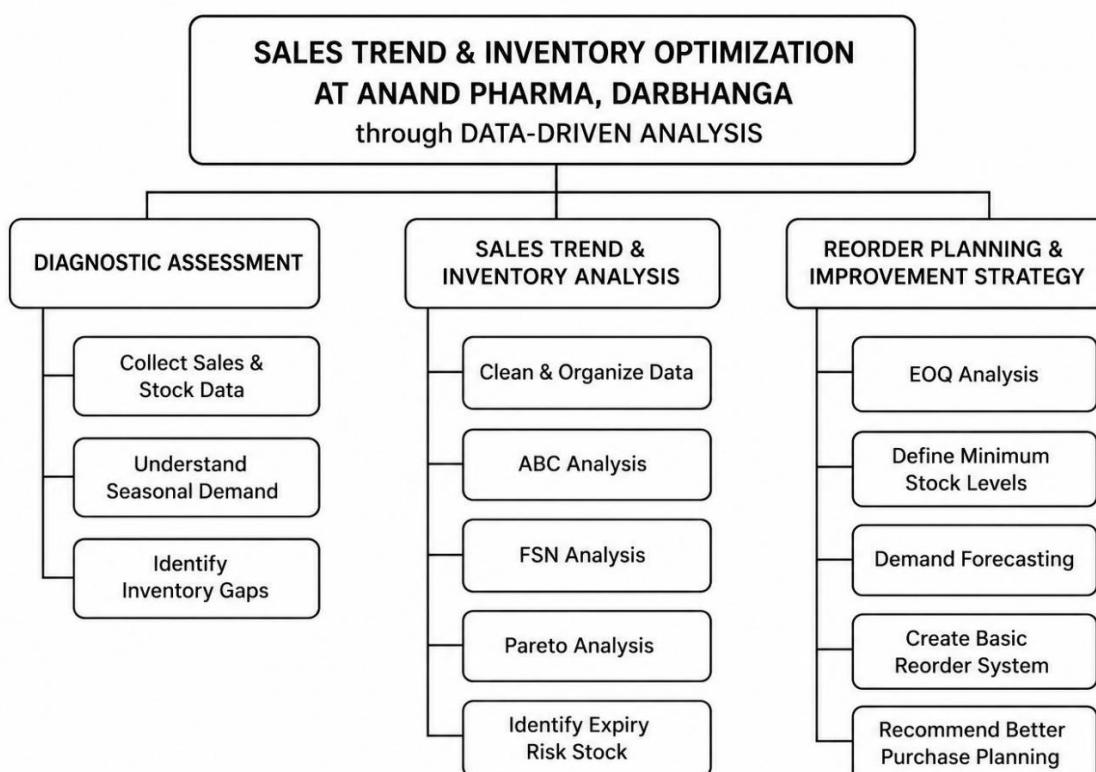
5.4.2 Demand Forecasting

Demand forecasting techniques such as Moving Average and Trend Analysis will be used to estimate future medicine demand. Forecasted demand will support better purchasing decisions and reduce the risk of stockouts and overstocking.

Overall, this step-by-step analytical approach will help Anand Pharma move from experience-based decisions to simple data-driven inventory planning, improving efficiency and financial stability.

6 Expected Timeline

6.1 Work Breakdown Structure:



6.2 Gantt Chart

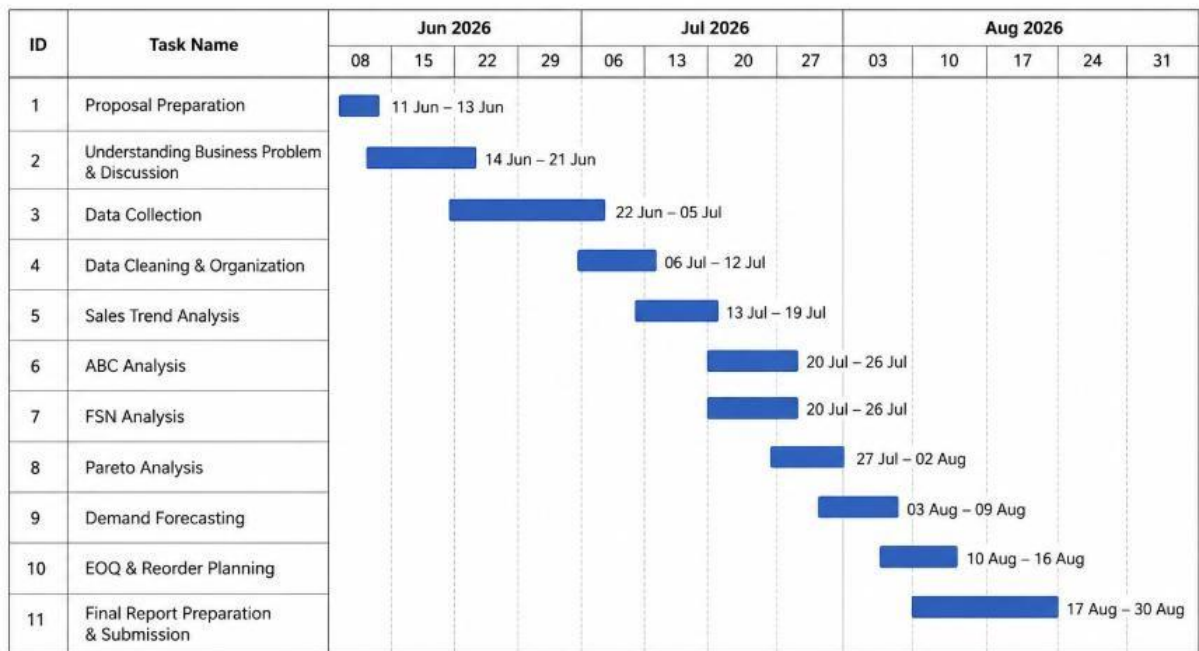


Figure 1: Expected Timeline for Completion of Project

7 Expected Outcome

7.1 Improved Inventory Planning

Through this proposal, Anand Pharma will be able to manage its stock in a more organized way. By studying which medicines sell quickly and which remain unsold, the pharmacy can avoid overstocking slow-moving medicines. It will also help ensure that essential medicines are always available when customers need them. Overall, this will reduce expiry losses and improve daily operations.

7.2 Better Cash Flow Management

With better inventory planning, the pharmacy will not have to block money in medicines that do not sell regularly. Instead, more focus can be given to high-

demand medicines. This will help in using money more wisely and improving overall cash flow, reducing unnecessary financial pressure.

7.3 Increased Customer Trust

When customers find the medicines they need without delay, they are more likely to return again. Proper stock availability and timely restocking will help Anand Pharma build stronger relationships with customers and maintain trust in the local community.

7.4 Basic Use of Digital Tools

This project will encourage the use of simple tools like Excel sheets for tracking sales and stock. Even basic digital records can reduce errors, save time, and give a clearer understanding of sales patterns.

7.5 Long-Term Business Stability

With a more structured system in place, the pharmacy will be better prepared for future growth. It can handle seasonal demand changes more confidently and continue serving the community effectively

Overall, the proposed study is expected to improve inventory visibility, support data-driven decision making, reduce expiry-related losses, and enhance the operational efficiency of Anand Pharma. The findings will help the pharmacy maintain optimum stock levels and improve long-term business sustainability.

